Faster Attacks on QR-UOV Signatures

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22 July 2026

Abstract

We give faster attacks on the QR-UOV signature scheme, a third-round candidate in NIST's Additional Digital Signature process. For the Round 2 Levels I, III, and V parameters, our principal equivalent-key attack costs about

$$2^{124.5}, \qquad 2^{177.8}, \qquad 2^{235.4},$$

Boolean gate operations—the same unit in which the security targets are stated, under a conversion that deliberately overcounts—against targets 2^{143}, 2^{207}, and 2^{272} and the submitters' reported 150, 211, and 277 bits.

The attack restricts the public quadratic forms to a random low-dimensional linear slice and solves the resulting square system. A well-behaved solution set contains exactly one point lying on the hidden oil space, and the derivatives of the equations at that point recover the entire oil space—enough to forge signatures. Where earlier work assumed the secret key avoided an exceptional set of unknown size, we prove that the attack succeeds for all but a negligible fraction of keys. All estimates use the specification's asymptotic cost model.

1 Introduction

Unbalanced Oil and Vinegar (UOV) signatures divide variables into vinegar and oil variables. In secret coordinates, oil variables never multiply one another. Equivalently, the public quadratic forms share a hidden linear subspace on which all associated bilinear forms vanish. Knowing any valid oil space is enough to sign: after choosing vinegar variables, the verification equations become linear in the oil variables [3].

QR-UOV compresses the UOV public key using quotient-ring, or extension-field, structure [4]. NIST advanced QR-UOV to the third round of its Additional Digital Signature process in May 2026 [5, 6]. The current Round 2 parameter sets are

$$(q, v, m, \ell) = (127, 156, 54, 3), \quad (127, 228, 78, 3), \quad (127, 306, 105, 3),$$

for Levels I, III, and V [1]. The submitters report best classical attacks of 150, 211, and 277 bits.

The specification exposes two public descriptions. First, it gives a smaller UOV tuple over

$$K = \mathbb{F}_{127^3}, \qquad V = v/3 \in \{52, 76, 102\}, \qquad O = m/3 \in \{18, 26, 35\}.$$

Second, the verifier is a base-field quadratic map with shapes $(n, m) = (210, 54), (306, 78), (411, 105)$. Our equivalent-key attack uses the first description; the secondary direct-forgery path uses the second.

1.1 Main attack

The public key is a list of m symmetric matrices $P_1, \ldots, P_m$ over K, acting on the space K^{V+O}. The secret structure is a hidden O-dimensional subspace $\mathcal{O}$, the *oil space*, on which all m forms vanish; recovering any such space is enough to forge signatures. The attacker picks a random $(V+1)$-dimensional slice of the ambient space, recorded as a random full-rank matrix $M \in L^{(V+O)\times(V+1)}$ over an extension field L/K (the field constructed by the attack's finite-field polynomial solver, described below). Since the slice has dimension $V+1$ and the oil space has dimension O inside a space of dimension $V+O$, their dimensions add to more than the ambient dimension, so the slice always meets the oil space.

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Restricting the m public forms to this slice and taking V random linear combinations gives a square system of V quadratic equations. When it is nondegenerate (a smooth complete intersection), it has exactly 2^V solutions. Exactly one of them lies on the oil space, because the slice meets the oil space in a single point; any larger overlap would produce a whole line of shared solutions. At that one solution, the derivatives of the equations vanish precisely along the oil directions, so the kernel of the Jacobian there is exactly the oil space $\mathcal{O}_L$. The attacker reads it off by row reduction, checks that it is defined over the base field K (Frobenius descent) and satisfies the public identities, and maps it back to an ordinary base-field signing key through the specification's inverse pull-back.

A more naive version of this idea uses a single *fixed* slice. There one can only show the solve is fast for keys outside some exceptional set, with no control on how large that set is; for a fixed slice this gap is real, because the bad set is large (it has low codimension), as we quantify in Appendix C. Randomizing the slice removes the problem: we prove the attack succeeds unless the key falls into one of three degenerate cases—no usable oil direction, an unwanted extra solution lying off the oil space, or a linear dependence among the forms. The dominant case has codimension

$$e(O+1)+1, \qquad e = m - V,$$

namely $39, 55, 109$ for the three levels, and a degree and point-count estimate turns this into the tiny bad-key probabilities quoted below.

1.2 Attack at a glance

The complete procedure is the following.

Fast equivalent-key recovery.

1. Expand the compressed public key to the public extension-field UOV tuple and construct the solver field L.
2. Sample $M \in L^{(V+O)\times(V+1)}$, reject unless $\operatorname{rank} M = V+1$, and restrict all m public quadrics to $\mathbb{P}(\operatorname{im} M)$.
3. Sample $R \in L^{V\times m}$, reject unless $\operatorname{rank} R = V$, and run the fast Kronecker solver on the V recombined restricted quadrics.
4. Retain L-rational roots satisfying all m unrecombined restricted equations. At each survivor, compute the ambient derivative kernel and accept only an O-space that descends to K and passes all public graph and total-isotropy checks.
5. Apply the public inverse pull-back to obtain an equivalent base-field signing key. On failure, resample (M, R).

Direct forgery. Under the published pseudo-oil extraction heuristic, restrict the full base-field map to a structured m-space, solve two univariate quadratics, eliminate one variable linearly, solve a final square system in $m-3$ variables, and verify the signature.

Verification makes both procedures one-sided: random choices affect termination, never acceptance of a false key or signature.

Theorem 1.1 (Fast almost-all-key recovery, informal)**.** *For Levels I, III, and V, the fast projective-slice attack performs about $2^{117.59}, 2^{170.39}, 2^{227.59}$ bit operations, equivalently a Boolean circuit of about $2^{124.47}, 2^{177.80}, 2^{235.42}$ gates—the unit in which the security targets are stated. Its ideal-field key-bad probability is below $2^{-364.56}, 2^{-457.92}, 2^{-1038.06}$. For a key outside that bad set, a single attempt—one random slice M and one random recombination R, over the solver fields already in use—fails the smoothness test with probability at most $2^{-145.65}, 2^{-183.46}, 2^{-219.51}$.*

Rigor and cost model. The algebraic reductions and probability bounds are exact where stated. Key-density calculations use the ideal-field experiment; fixed-budget verified-success probabilities transfer computationally to SHAKE/AES expansion. All costs are asymptotic bounds with large hidden constants and memory, not implementation forecasts. Pseudo-oil existence, extraction, and branch behavior use the cited average-case heuristic.

1.3 Related work

QR-UOV was introduced as a quotient-ring compression of UOV at ASIACRYPT 2021 [4]. The Round 2 specification analyzes direct MQ, reconciliation, intersection, PXL, and rectangular-MinRank attacks [1]. Furue and Ikematsu sharpened the rectangular-MinRank line, and Suzuki et al. extended that framework while

Table 1: Headline attacks. Entries count soft bit-operations; the parenthesized figures are a more conservative conversion to Boolean-gate-circuit size, the unit in which the security targets are stated.

Attack and output	Required condition	Level I	Level III	Level V
Fast projective-slice key recovery; equivalent signing key	Ideal-field theorem; transfer to concrete seeded keys costs a key-expansion distinguishing advantage plus a rejection-sampling term (Adv $+\varepsilon_{\mathrm{RS}}$)	117.59 (124.47)	170.39 (177.80)	227.59 (235.42)
Pseudo-oil direct forgery	Published extraction and branch heuristic	129.52 (136.53)	180.06 (187.55)	236.18 (244.06)
Nominal target	—	143	207	272

reporting that the recommended sets remained above target in their analyses [16, 17]. Our key recovery instead exploits the public extension-field tuple, projective slicing, complete polynomial-system solving, and derivative-kernel recovery.

Beullens' intersection attack and subsequent UOV cryptanalysis recover hidden oil geometry by structured linear algebra [15]. Pébereau showed that one suitable oil vector suffices for polynomial-time equivalent-key recovery [18]. The derivative-kernel theorem used here is a public-coordinate realization of that principle; the earlier one-column completion theorem is a QR-UOV-specific graph-coordinate specialization.

Van der Hoeven and Lecerf give the nearly quadratic finite-field Kronecker solver used by the main attack [8]. Benoist computes the complete-intersection discriminant degrees used in the regularization analysis [7]. The robustness appendix draws on geometric resolutions and symbolic homotopy due to Schost, Safey El Din–Schost, and Vu [10, 9, 11], with a finite-characteristic start system, separator, and dense-quadratic batching supplied here. Its field-uniform matrix-algebra ledger uses Strassen [14], and a separate cubic line verifies that the below-target conclusion does not rely on fast matrix multiplication.

May, Ostuzzi, and Ressler's *Just Guess* and Ostuzzi's enriched pseudo-oil framework give improved algorithms for underdetermined MQ [19, 20]. Relative to those algorithms, the new parts of our direct-forgery path are the feasibility obstruction that remains after slicing, the reversed order in which subspace extraction precedes geometric resolution, and the parameter audit specialized to QR-UOV. Jin et al. study odd-characteristic UOV families through symmetric algebra but report no stronger QR-UOV figures in that line [21]. Side-channel and vinegar-reuse issues are orthogonal and are surveyed by Aulbach, Campos, and Krämer [24].

Concurrent structural cryptanalysis. This paper is one of three concurrent studies by the same authors in a common program of structural cryptanalysis of UOV-derived signatures. The other works analyze MAYO/UOV using polar-flag quotients and Hodge-contraction acceleration, and odd-characteristic SNOVA using a symmetric-square quotient of the ordered power-pair feature space [22, 23]. None of the three papers is needed for the correctness of either of the others. They share a broader lesson: the public algebra used to compress or structure a UOV-derived construction must be incorporated explicitly into its security analysis rather than erased by modeling the resulting system as generic MQ.

2 The public key over the extension field

This section fixes the notation the attack works with. Recall from Section 1 that the specification exposes a compact UOV key over the extension field $K = \mathbb{F}_{127^3}$: a list of quadratic forms sharing a hidden oil space. We write that key out and record the recommended parameters. We then set up the probability model used for the key-density bounds—sampling the key material uniformly, rather than from the specification's hash-based expansion—and show in Proposition 2.2 how to transfer those bounds back to concrete seeded keys.

The public key-recovery tuple is

$$P = (p_1, \dots, p_m), \qquad p_i(x) = x^{\mathsf{T}} P_i x, \qquad x \in K^{V+O},$$

where P_i are symmetric and share a hidden totally isotropic O-space $\mathcal{O}$:

$$u^{\mathsf{T}} P_i v = 0 \qquad (u, v \in \mathcal{O}, 1 \le i \le m). \tag{1}$$

The recommended parameters are:

Level	q	ℓ	V	O	m
I	127	3	52	18	54
III	127	3	76	26	78
V	127	3	102	35	105

The same records appear in the active Round 2 reference implementation [2]. Throughout,

$$Q = |K| = 127^3 = 2{,}048{,}383, \qquad \log_2 Q = 20.966054\ldots.$$

2.1 Ideal-field model and concrete expansion

Definition 2.1 (Ideal-field experiment)**.** *Whenever the specification's public- or secret-key expansion routine (Expandpk or Expandsk) requests an element of $\mathbb{F}_q$, sample it independently and uniformly from $\mathbb{F}_q$, then run all remaining key-generation steps unchanged.*

In the block notation set up in (2) and (3) below, the public blocks and the secret graph are independent and uniform under this experiment; the central matrices A_i, B_i are therefore independent and uniform as well.

The specification uses finite SHAKE/AES buffers and rejection sampling. If a call exhausts its buffer it substitutes zero; the buffers are sized so that any single call exhausts with probability at most $2^{-\lambda}$, where $\lambda \in \{128, 192, 256\}$ is the category security parameter for Levels I, III, and V. With $2m+1$ such calls, put

$$\varepsilon_{\mathrm{RS}} = (2m+1)2^{-\lambda}.$$

Here $\mathrm{Adv}^{\mathrm{exp}}_{\textsf{QR-UOV}}(D)$ denotes the advantage of a distinguisher D in telling the scheme's real SHAKE/AES key-expansion output from independent uniform bytes. Bounding it is a standard pseudorandomness assumption on the expansion function, and every transfer below to concrete seeded keys is conditional on it; the ideal-field bounds themselves are unconditional.

Proposition 2.2 (Expansion transfer)**.** *Let U be any auxiliary public random variable sampled independently of key generation and identically in the concrete and ideal experiments. For any event E decidable from the expanded key-generation matrices and U, there is a distinguisher D_E, with the cost of key generation plus the event test, such that*

$$\left|\Pr_{\text{concrete}}[E] - \Pr_{\text{ideal}}[E]\right| \le \mathrm{Adv}^{\mathrm{exp}}_{\textsf{QR-UOV}}(D_E) + \varepsilon_{\mathrm{RS}}.$$

Proof. Condition on the value of U. Replace the concrete expansion transcript by independent uniform bytes, paying the transcript distinguishing advantage, and abort if any rejection buffer contains too few accepted bytes, paying at most $(2m+1)2^{-\lambda}$. Conditioned on no abort, the accepted values are independent and uniform and therefore give exactly Definition 2.1. Averaging over U proves the claim. In particular, U may contain the public slice, output-mixing, separator, and solver coins of any fixed-budget verified attack experiment. $\square$

The reported rejection-term logarithms are $-121.23, -184.71, -248.28$. We apply Proposition 2.2 to the public verified-success experiment, whose test cost is the attack cost, rather than to exact algebraic-locus membership.

2.2 Central and public graph coordinates

In secret central coordinates $K^V \oplus K^O$,

$$F_i = \begin{pmatrix} A_i & B_i \\ B_i^\top & 0 \end{pmatrix}, \qquad F_i(v,o) = v^\top A_i v + 2v^\top B_i o, \tag{2}$$

with $A_i \in \mathrm{Sym}_V(K)$ and $B_i \in K^{V\times O}$ independent uniform. The planted oil space is $\mathcal{O}_0 = \{0\} \oplus K^O$.

In normalized public coordinates the hidden space is a graph

$$\mathcal{O} = \left\{ \begin{pmatrix} -Gc \\ c \end{pmatrix} : c \in K^O \right\}. \tag{3}$$

Writing

$$P_i = \begin{pmatrix} A_i & E_i \\ E_i^\top & C_i \end{pmatrix},$$

total isotropy is equivalent to

$$C_i = G^\top E_i + E_i^\top G - G^\top A_i G. \tag{4}$$

Proposition 2.3 (Verified graph gives an equivalent key)**.** *Any $G' \in K^{V\times O}$ satisfying every identity (4) is converted by the specification's public inverse pull-back into an equivalent base-field* QR-UOV *signing key.*

Proof. Put

$$S_{G'} = \begin{pmatrix} I_V & G' \\ 0 & I_O \end{pmatrix}, \qquad S_{G'}^{-1} = \begin{pmatrix} I_V & -G' \\ 0 & I_O \end{pmatrix}.$$

A direct block multiplication gives

$$S_{G'}^{-\mathsf{T}} P_i S_{G'}^{-1} = \begin{pmatrix} A_i & E_i - A_i G' \\ E_i^{\mathsf{T}} - G'^{\mathsf{T}} A_i & C_i - G'^{\mathsf{T}} E_i - E_i^{\mathsf{T}} G' + G'^{\mathsf{T}} A_i G' \end{pmatrix}.$$

The lower-right block is zero exactly by (4), so these are valid central UOV forms and $S_{G'}$ is an equivalent secret linear map over K. The documented QR-UOV pull-back identifies the structured base-field blocks with their matrices over K and is publicly invertible on its image. Applying that inverse entrywise therefore gives a base-field secret key whose public forms are the original ones. Running the specification's signing algorithm with this key produces signatures accepted by the unchanged verifier. □

3 Recovery and structural certificates

Once the attack has a projective point x lying on the oil space, three things remain: read the full oil space off from x, certify and complete it into a signing key, and confirm that the recovered space is genuinely the hidden one. This section supplies each step—derivative-kernel recovery (Lemma 3.1), one-column graph completion (Theorem 3.2), and a uniqueness bound ruling out a second oil space (Theorem 3.5)—together with a 2^V optimality floor for the associated square systems (Theorem 3.6).

For a public common zero x, form

$$C_x = (P_1 x \ \cdots \ P_m x) \in L^{(V+O)\times m}.$$

Lemma 3.1 (Derivative-kernel recovery)**.** *If $x \in \mathbb{P}(\mathcal{O}_L)$, then $\mathcal{O}_L \subseteq \ker C_x^{\mathsf{T}}$. If $\operatorname{rank} C_x = V$, equality holds.*

Proof. For $u \in \mathcal{O}_L$, (1) gives $u^{\mathsf{T}} P_i x = 0$ for every i. If $\operatorname{rank} C_x = V$, its transpose kernel has dimension $(V+O) - V = O$. □

A smooth restricted square core has derivative rank V at its oil point, while total isotropy bounds the ambient derivative rank by V. Thus Lemma 3.1 recovers the complete oil space. Row reduction into the public graph chart, Frobenius descent to K, verification of (4), and Proposition 2.3 finish the attack.

3.1 One-column completion

Let $G = [g_1|\cdots|g_O]$. For a coordinate-slice candidate g, define

$$(M_j(g))_{i,*} = (A_i g - E_i e_j)^{\mathsf{T}}, \qquad (H_j(g))_{i,r} = (E_i e_r)^{\mathsf{T}} g - (C_i)_{jr}.$$

Theorem 3.2 (One column completes the graph)**.** *For the true column g_j,*

$$M_j(g_j) G_{-j} = H_j(g_j).$$

If $\operatorname{rank} M_j(g_j) = V$, all remaining columns are uniquely recovered by one public multi-right-hand-side solve.

Proof. The (j,r) entry of (4) rearranges to

$$(A_i g_j - E_i e_j)^{\mathsf{T}} g_r = (E_i e_r)^{\mathsf{T}} g_j - (C_i)_{jr}.$$

Stacking over i, r gives the matrix identity. Full column rank gives uniqueness. □

3.2 Generic uniqueness

We now check that the space the attack recovers is the true oil space: for almost every key there is no second common isotropic O-space that could be confused with it. Fix the planted oil space $\mathcal{O}_0$. Let $W \neq \mathcal{O}_0$ be another O-space and put

$$s = O - \dim(W \cap \mathcal{O}_0), \qquad 1 \le s \le O.$$

Lemma 3.3 (Relative Grassmannian stratum)**.** *The locus of such W with fixed s has dimension $s(V+O-s)$.*

Proof. Choose the $(O-s)$-dimensional intersection inside $\mathcal{O}_0$, the s-dimensional image of W in $K^{V+O}/\mathcal{O}_0 \simeq K^V$, and the graph map back to the s-dimensional quotient of $\mathcal{O}_0$. The dimensions are $(O-s)s$, $s(V-s)$, and s^2, respectively. □

Lemma 3.4 (Independent extra isotropy conditions)**.** *For fixed W in this stratum, requiring one central form that already vanishes on $\mathcal{O}_0 \times \mathcal{O}_0$ also to vanish on $W \times W$ imposes exactly*

$$c_s = \binom{O+1}{2} - \binom{O-s+1}{2} = \frac{s(2O-s+1)}{2}$$

additional independent linear conditions.

Proof. Restriction to W already vanishes on $\mathrm{Sym}^2(W \cap \mathcal{O}_0)$. Conversely, after choosing complements, every symmetric form on W that vanishes on this subspace extends to an ambient symmetric form vanishing on $\mathcal{O}_0 \times \mathcal{O}_0$. The restriction map therefore has the displayed rank. □

Theorem 3.5 (Uniqueness codimension and rational first moment)**.** *For m independent central forms, the key locus admitting an alternative common isotropic O-space of type s has codimension at least*

$$\delta_s = m\frac{s(2O-s+1)}{2} - s(V+O-s).$$

The minimum over s equals $903, 1927, 3539$ *for Levels I, III, and V. Moreover, with Gaussian binomial coefficients denoted by $\begin{bmatrix} a \\ b \end{bmatrix}_Q$,*

$$\Pr[\textit{a second } K\textit{-rational oil space}] \leq \sum_{s=1}^{O} \begin{bmatrix} O \\ s \end{bmatrix}_Q \begin{bmatrix} V \\ s \end{bmatrix}_Q Q^{s^2-mc_s}.$$

The base-two logarithms of this bound are $-18932.35, -40401.59, -74198.87$.

Proof. Over the stratum of dimension $s(V+O-s)$, each of the m form fibers loses c_s dimensions by Lemma 3.4. The incidence variety therefore has codimension at least $mc_s - s(V+O-s)$ after projection to key space. The function δ_s is concave in s, so its minimum on $1 \leq s \leq O$ occurs at an endpoint; direct evaluation gives the displayed values and shows that $s=1$ minimizes in all three parameter sets.

For the finite-field statement, the number of K-rational O-spaces in the type-s stratum is

$$\begin{bmatrix} O \\ s \end{bmatrix}_Q \begin{bmatrix} V \\ s \end{bmatrix}_Q Q^{s^2}.$$

For each fixed W, the m independent forms satisfy the c_s extra conditions with probability Q^{-mc_s}. The union bound gives the formula; exact evaluation gives the stated logarithms. □

3.3 Multihomogeneous path-count floor

For a square subsystem of the coupled graph equations, let d_j count selected diagonal equations in block j and c_{jk} selected cross equations. Its multihomogeneous Bézout number is

$$B = [t_1^V \cdots t_O^V] \prod_j (2t_j)^{d_j} \prod_{j<k} (t_j + t_k)^{c_{jk}}.$$

Theorem 3.6 (Bézout floor)**.** *If $B \neq 0$, then $B \geq 2^V$. Equality is attained by taking V diagonal equations in one root block and V cross equations on each edge of a rooted spanning tree. Thus one-column recovery followed by linear completion is optimal in multihomogeneous path count among square graph subsystems.*

Proof. Expand each cross factor by orienting a labeled edge toward the endpoint whose variable is selected. If $D = \sum_j d_j$, then $B = 2^D N$, where N counts orientations with indegrees $V - d_j$. If $D \geq V$, the claim is immediate. Otherwise every vertex has indegree at least $V - D$. Remove a directed cycle; after fewer than $V - D$ such removals, each vertex has lost at most one incoming edge per removed cycle and still has positive indegree, so another directed cycle remains. Hence there are $V - D$ edge-disjoint directed cycles. Reversing any subset preserves all indegrees and gives $N \geq 2^{V-D}$. The rooted-tree construction has one orientation and $D = V$. □

4 Random projective slicing and fast regularization

This section proves the regularization and key-density claims behind Theorem 1.1. We work in secret central coordinates, where the key distribution is transparent, and show that for all but a negligible fraction of keys a random slice together with a random recombination produces a smooth 2^V-point core from which the oil space is recovered.

Let $K = \mathbb{F}_{127^3}$, $Q = |K|$, and

$$(V, O, m) \in \{(52, 18, 54), (76, 26, 78), (102, 35, 105)\}.$$

Put $e = m - V \in \{2, 2, 3\}$ and $N = V + O$. Work first in secret central coordinates

$$W = K^V \oplus K^O, \qquad \mathcal{O}_0 = \{0\} \oplus K^O.$$

The m central quadrics are

$$F_i(v, o) = v^{\mathsf{T}} A_i v + 2v^{\mathsf{T}} B_i o, \qquad A_i \in \mathrm{Sym}_V(K),\ B_i \in K^{V\times O}. \tag{5}$$

In the ideal-field experiment, the pairs (A_i, B_i) are independent and uniform. Equivalently, each F_i is uniform in

$$\mathcal{U} := H^0\big(\mathbb{P}(W), \mathcal{I}_{\mathbb{P}(\mathcal{O}_0)}(2)\big), \qquad D_{\text{form}} := \dim \mathcal{U} = \frac{V(V+1)}{2} + VO.$$

The proof is written in central coordinates only to expose the distribution. The secret change from public to central coordinates lies in $\mathrm{GL}_N(K)$; left multiplication by its inverse is a bijection on $L^{N\times(V+1)}$. Thus a public uniform slice matrix remains uniform in central coordinates, while smoothness, intersection multiplicity, and derivative-kernel recovery are coordinate invariant. In particular, the public and central tuples have identically vanishing joint discriminant polynomials simultaneously. Let

$$X_F := V(F_1, \dots, F_m) \subset \mathbb{P}(W).$$

It contains the planted oil space $\mathbb{P}(\mathcal{O}_0) \simeq \mathbb{P}^{O-1}$.

The fast branch samples

$$M \leftarrow L^{N\times(V+1)}$$

and rejects unless $\operatorname{rank} M = V + 1$, where L/K is the working extension already constructed by the finite-field Kronecker solver. It then restricts the public quadrics to the projective V-plane $\Lambda_M = \mathbb{P}(\operatorname{im} M)$:

$$q_i(t) = F_i(Mt), \qquad t \in \mathbb{P}^V_L,$$

and then samples $R \leftarrow L^{V\times m}$, rejects unless $\operatorname{rank} R = V$, and solves

$$g_a(t) = \sum_{i=1}^{m} R_{a,i} q_i(t), \qquad 1 \le a \le V.$$

In secret affine coordinates, a transverse Λ_M is exactly a slice $o = c + Lv$ after translating its unique oil intersection. The projective parameterization is preferable in the proof because it has no inverses or denominators: every restricted coefficient is simply quadratic in the entries of M.

4.1 Why the full discriminant is enough

The van der Hoeven–Lecerf solver is stated under regular-sequence and absolute-radicality assumptions for all intermediate ideals. It is not necessary to force these assumptions by multiplying all prefix discriminants.

Lemma 4.1 (Full smooth core implies the solver's prefix hypotheses)**.** *Let $g_1, \dots, g_V$ be quadrics in $\mathbb{P}^V$ over a field of odd characteristic. If their common zero scheme consists of exactly 2^V distinct points and the projective Jacobian has rank V at every point, then the ordered system satisfies the regular-sequence and absolutely-radical intermediate-ideal hypotheses used by the Kronecker solver.*

Proof. Extend scalars to an algebraic closure. Choose a projective coordinate change whose affine chart contains all 2^V points, and choose a linear form separating them. Interpolation then gives a separable polynomial q of degree 2^V and coordinate polynomials $v_1, \dots, v_V$ parametrizing all roots. The original equations vanish modulo q, and the Jacobian determinant is invertible modulo q because every root is regular. These are exactly the hypotheses of the a posteriori criterion of van der Hoeven and Lecerf [8, Proposition 4.6]; it concludes that the sequence is regular and every intermediate ideal is absolutely radical. These properties are intrinsic and hence descend to the original field. The solver's remaining degree condition holds because every equation is quadratic. $\square$

Let $\Delta_{V,V}$ be the reduced discriminant for V quadrics in $\mathbb{P}^V$. It vanishes exactly when the common zero scheme is not a smooth complete intersection of codimension V. Benoist's degree formula [7] gives the partial degree

$$d_V = (V+1)2^{V-1} \tag{6}$$

in the coefficients of each equation. Thus it suffices to make the single polynomial

$$\Theta_F(M,R) := \Delta_{V,V}\left(\sum_i R_{1i}F_i(Mt), \dots, \sum_i R_{Vi}F_i(Mt)\right) \tag{7}$$

nonzero.

Lemma 4.2 (Nonvanishing certifies full parameter rank)**.** *If* $\Theta_F(M,R) \neq 0$, *then* $\operatorname{rank} M = V+1$ *and* $\operatorname{rank} R = V$.

Proof. If $\operatorname{rank} M < V+1$, then $\mathbb{P}(\ker M)$ is nonempty and every pull-back $F_i(Mt)$, together with all of its first derivatives, vanishes there; the core is singular. If $\operatorname{rank} R < V$, the output equations are linearly dependent and cannot cut a smooth codimension-V complete intersection. In either case the full discriminant vanishes. $\square$

4.2 Geometric nonvanishing of the joint slice-and-output discriminant

For $x \in X_F$, let

$$\rho_F(x) := \operatorname{rank}\bigl(dF_1(x), \dots, dF_m(x)\bigr),$$

where the differentials are taken on $T_x\mathbb{P}(W)$.

Definition 4.3 (Three key-level bad loci)**.** *Define:*

$$\begin{aligned}
\mathcal{B}_{\text{oil}} &:= \{F : \rho_F(p) < V \textit{ for every } p \in \mathbb{P}(\mathcal{O}_0)(\overline{K})\},\\
\mathcal{B}_{\text{off}} &:= \{F : \exists x \in X_F(\overline{K}) \setminus \mathbb{P}(\mathcal{O}_0) \textit{ with } \rho_F(x) < V\},\\
\mathcal{B}_{\text{dep}} &:= \{F : F_1, \dots, F_m \textit{ are linearly dependent in } \mathcal{U}\}.
\end{aligned}$$

The next three lemmas isolate the only geometry needed.

Lemma 4.4 (A transverse finite base slice through a regular oil point)**.** *Assume* $F \notin \mathcal{B}_{\text{oil}} \cup \mathcal{B}_{\text{off}}$. *Choose* $p \in \mathbb{P}(\mathcal{O}_0)(\overline{K})$ *with* $\rho_F(p) = V$. *Then a Zariski-open dense set of projective* V*-planes* $\Lambda \subset \mathbb{P}(W_{\overline{K}})$ *through* p *has all of the following properties:*

1. $\Lambda \cap \mathbb{P}(\mathcal{O}_0) = \{p\}$;
2. $X_F \cap \Lambda$ *is finite;*
3. *at every* $x \in X_F \cap \Lambda$, *the restricted differential* $T_x\Lambda \to \overline{K}^m$ *has rank* V.

Proof. First, $\dim X_F \leq O-1$. Otherwise a component of dimension at least O has a smooth point x outside $\mathbb{P}(\mathcal{O}_0)$, and then

$$\rho_F(x) \leq (N-1) - \dim T_x X_F \leq V-1,$$

contrary to $F \notin \mathcal{B}_{\text{off}}$. Since $\mathbb{P}(\mathcal{O}_0) \subset X_F$, the maximal dimension is exactly $O-1$.

At p, every differential annihilates $T_p\mathbb{P}(\mathcal{O}_0)$. Both this tangent space and $\ker dF(p)$ have dimension $O-1$, so they are equal. Therefore a general V-plane through p is transverse at p and intersects the oil space only in p. We henceforth restrict to this nonempty open subset of the Grassmannian; no other oil point can occur in the slice.

Let $\mathcal{G}_p$ be the Grassmannian of projective V-planes through p, of dimension $V(O-1)$. We now consider only $x \in X_F \setminus \mathbb{P}(\mathcal{O}_0)$. Put $K_x = \ker dF(x)$; the off-oil hypothesis gives $\dim K_x \leq O-1$. For fixed x, planes through p and x form a family of dimension $(V-1)(O-1)$. If the tangent direction of the line $\overline{px}$ is not in K_x, the additional condition $T_x\Lambda \cap K_x \neq 0$ is a proper Schubert condition and lowers this dimension by at least one. Allowing x to vary in dimension at most $O-1$ gives total dimension at most $V(O-1)-1$.

If the tangent direction of $\overline{px}$ lies in K_x, then every quadric F_i vanishes identically on that line: both endpoint values and the polar cross term vanish. The variety of such line directions through p has dimension at most $\dim X_F - 1 \leq O-2$. Planes containing one such line therefore again form a family of dimension at most

$$(O-2) + (V-1)(O-1) = V(O-1) - 1.$$

The off-oil rank-defect incidence is locally closed and has dimension at most $V(O-1)-1$ by the two cases above. Its closure in the projective product $X_F \times \mathcal{G}_p$ has the same dimension, and the projection of that closure is therefore a proper closed subset of $\mathcal{G}_p$. Outside it, every intersection point has restricted derivative rank V. A positive-dimensional intersection would have a smooth off-oil point with a nonzero tangent direction in $T_x\Lambda \cap \ker dF(x)$, a contradiction. Hence the intersection is finite. $\square$

Lemma 4.5 (A second-order spare equation forces separability)**.** *Let $f_1, \dots, f_m$ be quadrics on $\mathbb{P}^V$ with a common point p. Suppose their differential matrix at p has rank V. If there is a nonzero*

$$h \in \operatorname{span}(f_1, \dots, f_m)$$

whose first jet at p is zero, then the rational map

$$\phi_f : \mathbb{P}^V \dashrightarrow \mathbb{P}^{m-1}, \qquad x \longmapsto [f_1(x) : \cdots : f_m(x)]$$

has differential rank V on a nonempty open subset.

Proof. Choose projective coordinates with $p = [1 : 0 : \cdots : 0]$ and affine coordinates $y = (y_1, \dots, y_V)$. After an equation-basis change, the first V forms have expansions

$$f_j(y) = y_j + q_j(y),$$

where each q_j is quadratic. Since h is a quadric with zero value and zero differential at p, it is a nonzero homogeneous quadratic $q(y)$ on this chart.

Consider the $(V + 1) \times (V + 1)$ first-jet minor whose rows are $f_1, \dots, f_V, h$ and whose columns are value followed by the V first derivatives. Up to a nonzero sign, its lowest-degree homogeneous term is

$$q(y) - \nabla q(y) \cdot y = -q(y),$$

using Euler's identity $\nabla q(y) \cdot y = 2q(y)$. It is therefore nonzero. Hence the full first-jet rank is $V + 1$ generically, which is equivalent to differential rank V for ϕ_f away from the base locus. □

Lemma 4.6 (Finite regular base locus plus separability)**.** *Let $f_1, \dots, f_m$ be quadrics on $\mathbb{P}^V$ over an algebraically closed field. Assume:*

1. *their common base scheme $B = V(f_1, \dots, f_m)$ is finite and nonempty;*
2. *the $m \times V$ projective differential matrix has rank V at every point of B;*
3. *ϕ_f has differential rank V on a nonempty open subset.*

Then a nonempty Zariski-open set of matrices $R \in k^{V\times m}$ makes

$$\left(\sum_i R_{1i} f_i, \dots, \sum_i R_{Vi} f_i \right)$$

a smooth zero-dimensional complete intersection of degree 2^V.

Proof. Let $\mathcal{R} = k^{V\times m}$. For $x \notin B$, write

$$a(x) = (f_1(x), \dots, f_m(x)) \neq 0$$

and let $J(x)$ be the differential matrix. A matrix R has x as a root exactly when $Ra(x) = 0$, a codimension-V linear condition on $\mathcal{R}$.

On the open set where the first-jet matrix $[a(x) \mid J(x)]$ has rank $V + 1$, the map

$$a(x)^{\perp} \longrightarrow T^*_x\mathbb{P}^V, \qquad r \longmapsto rJ(x)$$

is surjective. Consequently the additional condition $\det(RJ(x)) = 0$ has codimension one inside $Ra(x) = 0$. After allowing x to vary in a V-dimensional space, this singular-root incidence still has codimension at least one in $\mathcal{R}$.

The first-jet rank-defect locus away from B has dimension at most $V - 1$ by assumption 3. There the root condition $Ra(x) = 0$ alone has codimension V, so its total incidence also has dimension at most $\dim \mathcal{R} - 1$.

Finally, at each $x \in B$, the root condition is automatic, but the rank-V derivative assumption makes $\det(RJ(x)) = 0$ a genuine hypersurface in $\mathcal{R}$. Since B is finite, these basepoint hypersurfaces do not fill $\mathcal{R}$.

Globally, the singular-root incidence is the closed subset of $\mathbb{P}^V \times \mathcal{R}$ cut out by $Ra(x) = 0$ and the maximal minors of $RJ(x)$. Its projection is closed because $\mathbb{P}^V$ is proper, and the preceding stratified dimension bounds show that this projection is proper. Choose R outside it. The resulting projective zero scheme is nonempty because it contains B. It cannot have positive dimension: a positive-dimensional component has a smooth point at which the Jacobian rank is below V. Therefore the zero scheme is smooth and zero-dimensional. Since it is cut out by V quadrics in $\mathbb{P}^V$, it is a complete intersection of degree 2^V. □

Theorem 4.7 (The enlarged exceptional locus is explicitly controlled)**.** *If*

$$F \notin \mathcal{B}_{\text{oil}} \cup \mathcal{B}_{\text{off}} \cup \mathcal{B}_{\text{dep}},$$

then the joint polynomial $\Theta_F(M, R)$ in (7) is nonzero.

Proof. Choose a regular oil point p. By Lemma 4.4, a dense open subset of the irreducible Grassmannian of V-planes through p has finite base intersection and full restricted derivative rank at every basepoint.

The coefficient kernel of the derivative map at p has dimension $m - V = e > 0$. Because $F \notin \mathcal{B}_{\text{dep}}$, any nonzero vector in this kernel produces a nonzero global quadric $H \in \text{span}(F_i)$ with zero first jet at p. The condition $H|_\Lambda \not\equiv 0$ is open and nonempty: if H vanished on every V-plane through p, it would vanish on their union, which is all of $\mathbb{P}(W)$. Intersecting the two nonempty open subsets gives a plane Λ satisfying both properties.

On this plane, Lemma 4.5 makes the restricted linear system generically separable, and Lemma 4.6 supplies an output matrix R whose full core is a smooth complete intersection of degree 2^V. Thus the discriminant evaluates nontrivially at one pair (M, R), proving that Θ_F is not the zero polynomial. □

4.3 Finite-field density of the three bad loci

4.3.1 First-jet surjectivity away from oil

Lemma 4.8 (The central-form space spans every off-oil first jet)**.** *For every $x \in \mathbb{P}(W) \setminus \mathbb{P}(\mathcal{O}_0)$, the first-jet evaluation map*

$$\mathcal{U} \longrightarrow K \oplus T_x^*\mathbb{P}(W)$$

is surjective after scalar extension to $\overline{K}$.

Proof. Choose a vinegar coordinate v_j that is nonzero at x and normalize to the chart $v_j = 1$. The quadrics

$$v_j^2, \quad v_j v_k \ (k \neq j), \quad v_j o_a \ (1 \leq a \leq O)$$

belong to $\mathcal{U}$ and restrict to the affine functions 1, the remaining vinegar coordinates, and the oil coordinates. Their first jets form a basis of the value-and-cotangent space. □

4.3.2 The off-oil incidence

Let $\mathcal{A} = \mathcal{U}^m$ be the affine key space. For an affine variety, write cdeg for the sum of the degrees of its irreducible components. For each vinegar chart $C_j \simeq \mathbb{A}^{N-1}$, let $Z_j \subset \mathcal{A} \times C_j$ be the incidence of pairs (F, x) such that all $F_i(x)$ vanish and the $m \times (N - 1)$ derivative matrix has rank at most $V - 1$.

Lemma 4.9 (Codimension and cumulative degree of the off-oil incidence)**.** *Each Z_j is pure of codimension*

$$m + c_{\text{det}}, \qquad c_{\text{det}} = (m - V + 1)O = (e + 1)O,$$

and satisfies

$$\text{cdeg}(Z_j) \leq 3^m(2V)^{c_{\text{det}}}.$$

If Y_j is the Zariski closure of its projection to key space, then

$$\text{codim}_{\mathcal{A}} Y_j \geq c_{\text{off}} := m + c_{\text{det}} - (N - 1) = e(O + 1) + 1, \qquad \text{cdeg}(Y_j) \leq \text{cdeg}(Z_j).$$

Proof. On the chart $v_j = 1$, the fixed monomials v_j^2, $v_j v_k$ for $k \neq j$, and $v_j o_a$ restrict to the affine functions 1, the remaining vinegar coordinates, and the oil coordinates. They therefore give a polynomial splitting of the value-and-first-jet evaluation map. After a triangular polynomial change of coefficient coordinates, the incidence is the product of an affine space, the equations setting m value coordinates to zero, and the determinantal variety of $m \times (N - 1)$ matrices of rank at most $V - 1$. It is consequently pure of the stated codimension.

In the original joint coefficient and point coordinates, each value equation has total degree at most three, each derivative entry has degree at most two, and every $V \times V$ minor has degree at most $2V$. The splitting above shows that Z_j is irreducible and that the value equations and determinantal conditions have the asserted total codimension. Over the algebraic closure, choose c_{det} generic linear combinations of the maximal minors so that, after the m value equations, they cut properly and retain Z_j as an irreducible component. Cumulative Bézout therefore gives

$$\text{cdeg}(Z_j) \leq 3^m(2V)^{c_{\text{det}}}$$

by the cumulative-degree theorem of Heintz and Lachaud–Rolland [25, 26].

Projection cannot increase dimension, so the projected codimension is at least the source codimension minus the chart dimension $N-1$. For the degree statement, let Y be an irreducible component of Y_j and choose an irreducible component Z of Z_j that dominates it. Intersect Z with $\dim Z - \dim Y$ general affine hyperplanes so that the resulting Z' still dominates Y generically finitely and $\deg Z' \le \deg Z$. A general complementary linear section of Y pulls back to at least $\deg Y$ points of Z', counted with multiplicity, while their number is at most $\deg Z'$. Thus $\deg Y \le \deg Z$; summing over image components gives $\mathrm{cdeg}(Y_j) \le \mathrm{cdeg}(Z_j)$. □

Proposition 4.10 (Explicit off-oil density)**.** *Under the ideal-field distribution,*

$$\Pr[F \in \mathcal{B}_{\mathrm{off}}] \le \eta_{\mathrm{off}} := V\,3^m(2V)^{(e+1)O}Q^{-\{e(O+1)+1\}}. \tag{8}$$

Proof. Every off-oil point lies in one of the V vinegar charts, so $\mathcal{B}_{\mathrm{off}} \subseteq \bigcup_j Y_j$. The affine finite-field point bound

$$|Y_j(\mathbb{F}_Q)| \le \mathrm{cdeg}(Y_j)Q^{\dim Y_j}$$

from Lachaud–Rolland [26], followed by the union bound and Lemma 4.9, gives the formula after division by $|\mathcal{A}(\mathbb{F}_Q)|$. □

4.3.3 The oil and dependence loci

Let

$$\tau_{m,V} = 1 - \prod_{j=0}^{V-1}(1-Q^{j-m})$$

be the rank-deficiency probability of a uniform $m \times V$ matrix. If every geometric oil direction is rank deficient, then in particular all O standard oil-coordinate directions are rank deficient. Their derivative matrices are independent in the ideal-field experiment. Hence

$$\Pr[F \in \mathcal{B}_{\mathrm{oil}}] \le \eta_{\mathrm{oil}} := \tau_{m,V}^{O}. \tag{9}$$

The m central forms are independent uniform vectors in a D_{form}-dimensional space. Hence

$$\Pr[F \in \mathcal{B}_{\mathrm{dep}}] = 1 - \prod_{j=0}^{m-1}(1-Q^{j-D_{\mathrm{form}}}) \le \eta_{\mathrm{dep}} := Q^{-D_{\mathrm{form}}}\frac{Q^m-1}{Q-1}. \tag{10}$$

Combining Theorem 4.7 with (8), (9), and (10) gives the promised key theorem.

Theorem 4.11 (Almost-all-key joint regularizability)**.** *In the ideal-field experiment,*

$$\Pr[\Theta_F \equiv 0] \le \eta_{\mathrm{off}} + \eta_{\mathrm{oil}} + \eta_{\mathrm{dep}}. \tag{11}$$

For the recommended parameters:

Level	c_{off}	$\log_2 \eta_{\mathrm{off}}$	$\log_2 \eta_{\mathrm{oil}}$	$\log_2 \Pr[\Theta_F \equiv 0]$
I	*39*	-364.564	-1132.167	< -364.563
III	*55*	-457.920	-1635.352	< -457.919
V	*109*	-1038.067	-2935.248	< -1038.066

The dependence term is below $2^{-4.7\times10^4}$ *already at Level I and is invisible at the displayed precision.*

Proof. By Theorem 4.7, the event $\Theta_F \equiv 0$ is contained in $\mathcal{B}_{\mathrm{oil}} \cup \mathcal{B}_{\mathrm{off}} \cup \mathcal{B}_{\mathrm{dep}}$. Apply (8), (9), (10), and the union bound. The displayed values are exact evaluations of those formulas. □

4.4 One-trial regularization probability in the existing solver field

Let d_V be as in (6). A coefficient of $F_i(Mt)$ is quadratic in M, while the ath combined equation is linear in the ath row of R. Since $\Delta_{V,V}$ has degree d_V in each equation, the joint polynomial obeys

$$\deg_M \Theta_F \le 2Vd_V, \qquad \deg_R \Theta_F \le Vd_V, \qquad \deg \Theta_F \le 3V(V+1)2^{V-1}. \tag{12}$$

Thus Schwartz–Zippel gives:

Corollary 4.12 (Joint slice-and-output failure). *For every key outside the bad set of Theorem 4.11, if M and R are uniform over a finite extension L/K, then*

$$\Pr[\Theta_F(M,R)=0] \le \delta_V(L) := \frac{3V(V+1)2^{V-1}}{|L|}. \tag{13}$$

For the existing working-field degrees $[L:K]=10,13,16$:

Level	$[L:K]$	$\log_2 \deg \Theta_F$	$\log_2 \delta_V(L)$	$\log_2 \delta_V(L)^2$
I	*10*	64.013	-145.647	-291.294
III	*13*	89.100	-183.459	-366.918
V	*16*	115.944	-219.513	-439.026

Proof. For a nonexceptional key, Θ_F is a nonzero polynomial by Theorem 4.11. Its total degree is bounded by (12), so Schwartz–Zippel gives (13). Rank-deficient M or R already lie in its zero set by Lemma 4.2; no separate rejection term is needed. ☐

A smooth full core has all 2^V roots distinct and regular, so Lemma 4.1 supplies the solver's regularity hypotheses. The displayed probabilities concern only the public slice-and-output regularization; the solver's own random choices retain the independently verifiable fixed-failure or Las Vegas guarantees of van der Hoeven–Lecerf. The two-trial column is not needed for expected-time complexity; it is useful when one wants the algebraic regularization failure of a fixed bounded-trial experiment to be far below every category target. Including the key-bad probability of Theorem 4.11 does not change any displayed exponent.

4.4.1 Fast-solver cost specialization

For quadrics, the Bézout number is $D=2^V$. Specializing the finite-field bound of van der Hoeven–Lecerf with their fixed parameter $\varepsilon>0$ gives the direct indicator

$$C_\varepsilon(V) = V + (1+\varepsilon)(V-1) + \log_2(3\log_2 127). \tag{14}$$

The finite-field extension construction is already included in this bound; sampling M and R in that same field adds only polynomially many field operations. The direct bit-operation indicator is $C_\varepsilon(V)$; the uniform Boolean-circuit indicator adds a $\log T$ factor, $C_\varepsilon(V)+\log_2 C_\varepsilon(V)$. At $\varepsilon=0.2$ the direct values are $117.59, 170.39, 227.59$ and the uniform values $124.47, 177.80, 235.42$. Because $\delta_V(L)$ is negligible, the expected number of public regularization trials on every nonexceptional key is $(1-\delta_V(L))^{-1}=1+o(1)$; the solver's independently verifiable internal random choices contribute only another constant factor.

4.5 End-to-end recovery from an extension-field projective slice

The new slice does not reveal a coordinate column of the graph matrix. Instead it gives a general oil vector, for which derivative-kernel recovery is stronger.

Lemma 4.13 (Recovery and descent from the projective oil point). *Let L/K be any finite extension. Suppose $\Theta_F(M,R)\neq 0$, and let*

$$[t^\star] \in \mathbb{P}^V(L)$$

be the unique point for which $x^\star = Mt^\star$ lies in the planted oil space $\mathbb{P}(\mathcal{O}_0)_L$. Put

$$C_{x^\star} = \begin{pmatrix} P_1x^\star & \cdots & P_mx^\star \end{pmatrix} \in L^{N\times m}.$$

Then

$$\operatorname{rank} C_{x^\star} = V, \qquad \ker C_{x^\star}^{\mathsf{T}} = \mathcal{O}_L.$$

The attacker can row-reduce this kernel into the public graph chart; the resulting graph matrix has entries in K, is fixed by the Q-Frobenius, and yields an equivalent base-field signing key after the public inverse pull-back.

Proof. By Lemma 4.2, M and R have full rank. Two L-linear subspaces of vector dimensions $V+1$ and O in an $N=V+O$ dimensional space always meet nontrivially. If their intersection had vector dimension at least two, every restricted output quadric would vanish on a positive-dimensional projective oil intersection, contradicting $\Theta_F(M,R)\neq 0$. Thus the oil intersection is a unique L-rational projective point.

Smoothness of the restricted core at $t^\star$ gives ambient derivative rank at least V. Total isotropy gives

$$\mathcal{O}_L \subseteq \ker C_{x^\star}^{\mathsf{T}},$$

so $\operatorname{rank} C_{x^\star} \le N - O = V$. Equality follows, and the two O-dimensional spaces coincide. The true public oil space is defined over K and transverse to the normalized public oil-coordinate factor, so its graph matrix recovered over L is automatically Q-Frobenius fixed. Direct verification of the graph identities and total isotropy makes the procedure Las Vegas. □

Root extraction. After undoing the solver's public coordinate change, the geometric resolution is defined over L and has a separable separator polynomial $q(T)$ of degree 2^V. Reduce $T^{|L|} - T$ modulo q and take the gcd with q. Because the separator is injective on the geometric roots and is defined over L, a root has separator value in L if and only if the entire point is fixed by $\mathrm{Gal}(\overline{L}/L)$; thus this gcd selects exactly the L-rational core roots. Evaluate each of the unrecombined $m - V$ equations in the quotient algebra and take successive gcds, factor the remaining polynomial over L, and reconstruct the corresponding coordinates. Fast modular exponentiation, gcd, factorization, and derivative-kernel linear algebra cost $\widetilde{O}(2^V \operatorname{poly}(V, O, \log |L|))$ field operations, below the nearly quadratic-in-2^V solver bound. Every survivor is checked against all original forms, so filtering errors cannot produce a false key.

4.6 Attack theorem and costs

Theorem 4.14 (Fast almost-all-key equivalent-key recovery)**.** *For each recommended parameter set, consider the following public trial over the fast solver's existing working field L:*

1. *sample M, reject unless* $\operatorname{rank} M = V + 1$*, and restrict all public extension-field quadrics to* $\mathbb{P}(\operatorname{im} M)$*;*
2. *sample R, reject unless* $\operatorname{rank} R = V$*, and run a bounded verified invocation of the fast Kronecker solver on the V combined restricted quadrics; failed inversions, a missing a posteriori certificate, or budget exhaustion return* FAIL*;*
3. *extract the L-rational roots satisfying all m unrecombined restricted equations;*
4. *at each survivor, compute the ambient derivative kernel and require dimension O, total isotropy, Frobenius descent to K, and every public graph identity;*
5. *apply the specification's public inverse pull-back to a verified graph.*

In the ideal-field experiment, the set of keys on which no pair (M, R) gives a smooth full core has probability below

$$2^{-364.56}, \quad 2^{-457.92}, \quad 2^{-1038.06}.$$

For every key outside that set, the public slice-and-output regularization part of one trial fails with probability at most

$$2^{-145.65}, \quad 2^{-183.46}, \quad 2^{-219.51}.$$

Because every public trial uses a bounded solver invocation and aborts on a missing certificate, repeated verified trials are Las Vegas on every such key: they eventually return an equivalent signing key with probability one and never return a false key. On an arbitrary key, any fixed-budget version is one-sided and may report failure.

The restriction, rational-root filtering, derivative-kernel recovery, and descent are lower order than the fast solve. The expected direct indicators (with uniform indicators in parentheses) on every nonexceptional key are therefore

$$117.59\ (124.47), \qquad 170.39\ (177.80), \qquad 227.59\ (235.42).$$

Two independent public regularization trials have the squared algebraic failure bounds in Corollary 4.12; solver-internal failure is separately amplified and verified at constant-factor cost.

Proof. The key-level exceptional probability is Theorem 4.11. On every nonexceptional key, Corollary 4.12 bounds the probability that the sampled pair (M, R) fails to give a smooth full core. Conditional on a smooth core, Lemma 4.1 supplies the hypotheses of the van der Hoeven–Lecerf solver, whose a posteriori check makes its repeated version Las Vegas. The unique projective oil intersection, derivative-kernel recovery, descent, and equivalent-key conversion are Lemma 4.13 and Proposition 2.3. Root extraction and every final acceptance test are performed against the unrecombined public equations, so no incorrect key is accepted.

The restriction and verification stages use only polynomially many operations beyond univariate arithmetic of degree 2^V. The root-filtering operations are quasi-linear, up to polynomial factors, in that degree and are therefore lower order than the solver bound. Substituting $D = 2^V$, degree two, and $\varepsilon = 0.2$ into (14) gives the direct indicators; adding the uniform $T \log T$ conversion gives the second numbers. The expected public retry factor is $(1 - \delta_V(L))^{-1} = 1 + o(1)$, and the solver's verified internal retry contributes only a constant factor. □

Corollary 4.15 (Fixed-budget concrete success)**.** *Let $t \ge 1$, and run t independent public slice-and-output trials. Amplify the verified internal solver so that the probability that all otherwise-good trials fail internally is at most β_t. In the ideal-field experiment, the probability of no verified equivalent key is at most*

$$\eta_{\mathrm{off}} + \eta_{\mathrm{oil}} + \eta_{\mathrm{dep}} + \delta_V(L)^t + \beta_t.$$

For the concrete SHAKE/AES-expanded distribution, the same failure probability is at most the displayed quantity plus

$$\mathrm{Adv}^{\mathrm{exp}}_{\textsf{QR-UOV}}(D_t) + \varepsilon_{\mathrm{RS}},$$

where D_t is the fixed-budget verified-success distinguisher and has the cost of those t attack trials plus key generation.

Proof. Split on the event $\Theta_F \equiv 0$. Its probability is bounded by Theorem 4.11. Outside it, independence of the public trial coins and Corollary 4.12 bound simultaneous algebraic regularization failure by $\delta_V(L)^t$; add the amplified internal failure β_t. The concrete statement is Proposition 2.2 applied to the verified-success event with all public coins included in U. □

4.7 Exact small-field audit of the new recovery chain

The accompanying script `qruov_projective_slice_toy_validation.py` implements the new slice and recovery chain over small odd prime fields at $V = 2$. For each sampled core it checks geometric smoothness exactly over the algebraic closure: on every affine chart of $\mathbb{P}^2$, it computes a Gröbner basis of the two restricted conics together with their affine Jacobian determinant. A unit ideal on every chart certifies that the projective core has no geometric singular root. It then computes the linear plane–oil intersection and verifies that the ambient derivative kernel equals the entire planted oil space.

The frozen audit suite covers shapes $(O, m) = (2, 4), (3, 4), (2, 3)$ over $\mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}$. Across 47 keys and 510 slice-and-output trials, 347 cores were geometrically smooth; every one of those 347 recovered the exact planted oil space. This is a controlled identity test, not an extrapolation to production dimensions, but it exercises precisely the projective-intersection and derivative-kernel steps newly introduced here.

Concrete seed expansion. The fixed-budget statement is formalized in Corollary 4.15. The rejection-term logarithms are $-121.23, -184.71, -248.28$; this term exceeds the one-trial geometric bound only at Level I. It is a coupling loss in the ideal-to-concrete reduction and is therefore displayed separately from the geometric failure probabilities.

5 A pseudo-oil and geometric-resolution direct-forgery path

The projective-slice attack recovers an equivalent key from the public extension-field tuple. A logically independent attack acts on the full base-field verification map

$$P : \mathbb{F}_{127}^n \longrightarrow \mathbb{F}_{127}^m, \qquad (n, m) = (210, 54), (306, 78), (411, 105).$$

It uses the enriched pseudo-oil framework of May–Ostuzzi–Ressler and Ostuzzi [19, 20]. These methods guess a structured low-dimensional subspace on which part of the system linearizes, reducing forgery to a smaller square polynomial system. We retain this path because it constrains a different parameter surface: its terminal square-system dimension is governed mainly by m, whereas the fast equivalent-key attack is governed mainly by $V = v/3$.

5.1 Feasibility and the obstruction after slicing

An enriched pseudo-oil decomposition is parameterized by a guess dimension k, a number ℓ_{po} of linearized equations, positive block sizes $b_1, \ldots, b_p$, and a residual dimension $r \geq \ell_{\mathrm{po}}$. Put

$$B = \sum_{j=1}^{p} b_j, \qquad L_{\mathrm{po}} = B + \ell_{\mathrm{po}}, \qquad r = m - B - k.$$

The two feasibility inequalities in the adopted average-case model are

$$n - (r-1)(L_{\mathrm{po}} + 1) \geq L_{\mathrm{po}}, \tag{15}$$

$$n - m + 1 \geq \sum_{j=1}^{p} b_j \left(m - \sum_{i=1}^{j} b_i - k \right). \tag{16}$$

Theorem 5.1 (Post-slice pseudo-oil obstruction)**.** *For an enriched pseudo-oil decomposition with $\ell_{\mathrm{po}} \geq 1$:*
1. if $n < m$, no parameters satisfy (16);

2. if $n = m$, feasibility forces $B = \ell_{\text{po}} = 1$, $p = 1$, $b_1 = 1$, and $k = m - 2$.

Thus an overdetermined projective or affine QR-UOV slice admits no useful enriched pseudo-oil step, while a square slice admits only a degenerate q^{m-2} guessing attack.

Proof. Every factor in the right side of (16) is at least $r \geq \ell_{\text{po}} \geq 1$, so the right side is at least $B\ell_{\text{po}} > 0$. If $n < m$, the left side is nonpositive. If $n = m$, the left side is one, forcing $B = \ell_{\text{po}} = 1$ and hence $p = b_1 = 1$; equality then gives $m - b_1 - k = 1$. □

The extension-field key-recovery tuple is underdetermined, but not enough to help this framework: (16) implies $Br \leq n - m + 1$ and therefore $k \geq 2m - n - 2$. At the recommended shapes this means at least $36, 52, 71$ guessed extension-field variables, costing roughly $2^{755}, 2^{1090}, 2^{1489}$ before the terminal solve. The pseudo-oil step must therefore precede slicing and act on the full base-field map.

5.2 The working pre-slice construction

We now fix a concrete feasible decomposition. Choose

$$p = 2, \qquad (b_1, b_2) = (1, 1), \qquad \ell_{\text{po}} = 1, \qquad k = 0.$$

Then $B = 2$, $L_{\text{po}} = 3$, and $r = m-2$. For the three recommended shapes, the left side of (15) is respectively $6, 6, 3$, while the right side is 3; (16) has slack $52, 76, 100$. Thus the adopted heuristic predicts a valid decomposition at every level. The subspace-extraction step (called *centrifugation* in the pseudo-oil framework [19, 20]) constructs

$$U = U_0 \oplus U_1 \oplus \widetilde{U}, \qquad \dim U_0 = m - 2, \quad \dim U_1 = \dim \widetilde{U} = 1,$$

with the following pattern for the first three equations:

(a) U_0 is totally isotropic for equations $1, 2, 3$;
(b) $U_0 \oplus U_1$ is totally isotropic for equation 1;
(c) U_0 is orthogonal to $U_1 \oplus \widetilde{U}$ for equations $1, 2$;
(d) $U_0 \oplus U_1$ is orthogonal to $\widetilde{U}$ for equation 1.

Theorem 5.2 (Exact residual reduction)**.** *Conditioned on a successful subspace extraction with the parameters above, direct forgery reduces exactly to:*

1. *one univariate quadratic in the coordinate of $\widetilde{U}$;*
2. *one univariate quadratic in the coordinate of U_1;*
3. *one affine-linear equation eliminating one U_0 coordinate;*
4. *$m - 3$ quadratic equations in $m - 3$ variables.*

A solution of the final square system reconstructs a preimage of the original verification target by linear algebra.

Proof. For equation one, $U_0 \oplus U_1$ is totally isotropic and orthogonal to $\widetilde{U}$, so after fixing the target the equation is univariate quadratic in the coordinate of $\widetilde{U}$. For equation two, U_0 is both totally isotropic and orthogonal to $U_1 \oplus \widetilde{U}$; after the first coordinate is fixed, the equation is univariate quadratic in the coordinate of U_1. For equation three, total isotropy of U_0 removes every U_0–U_0 term, so after the two one-dimensional coordinates are fixed the equation is affine-linear on U_0. Eliminating one coordinate leaves the final $m - 3$ equations on an affine space of the same dimension. Every step is an exact substitution, so reversing the substitutions reconstructs a preimage. □

Under the published extraction and branch heuristic, the two univariate stages generate one partial branch in expectation and the terminal residual is modeled as a generic square system with one nonsingular isolated solution in expectation. Failed or singular branches can be discarded, the decomposition rerandomized, or the public salt changed. These are the same average-case assumptions as in the underlying pseudo-oil analyses; the algebraic reduction after a valid decomposition is exact.

5.3 Costs

The primary pseudo-oil line uses the field-uniform arbitrary-core solver from Appendix B on the final $(m - 3)$-variable system and therefore inherits the preceding nonsingular-terminal-branch assumption. For the selected two one-dimensional blocks, the charged subspace-extraction work is $(m - 2)\,\mathrm{Solve}(3, 3) + \mathrm{Solve}(1, 1)$ plus polynomial-time linear algebra; the constant-size branch tree is also lower order than the terminal solve in the adopted model.

Table 2: Pseudo-oil direct forgery. Entries are direct indicators, with uniform indicators in parentheses.

Level	Final	Field-uniform Strassen	Target
I	51	129.52 (136.53)	143
III	75	180.06 (187.55)	207
V	102	236.18 (244.06)	272

6 Security consequences, cost conventions, and limitations

6.1 Cost conventions

The QR-UOV specification assigns security categories by counting the Boolean gates in a circuit that carries out the attack [1], so all comparisons must ultimately be in that unit. Our *direct* column is the number of bit operations the attack performs, read off from the cited solver theorems with quasi-linear arithmetic in the relevant finite field. Our *uniform* column converts that into a gate count: a T-step bit algorithm compiles into a Boolean circuit of size $O(T \log T)$ [13], so the uniform value is the direct value plus a $\log_2$ term. The uniform column is the one that matches the specification's unit, and it carries the main comparison.

Random choices are circuit inputs. A constant number of attempts can be run in parallel, every candidate can be checked publicly, and a selector can return the first valid output. Every cost is asymptotic and omits hidden constants, memory traffic, and wall-clock engineering.

6.2 Primary ledger

Table 3: Master security ledger. Entries are direct indicators, with uniform $T \log T$ indicators in parentheses. The arbitrary-core fallback is deliberately omitted from the primary comparison and reported in Table 4.

Level	Target	Spec. best	Fast key	Pseudo-oil	Fast uniform margin
I	143	150	117.59 (124.47)	129.52 (136.53)	18.53
III	207	211	170.39 (177.80)	180.06 (187.55)	29.20
V	272	277	227.59 (235.42)	236.18 (244.06)	36.58

The fast key-recovery line now comes with an explicit bound on the fraction of keys it can fail on, rather than merely holding outside an exceptional set of unquantified size. Its key-level exceptional probabilities are far below the category targets in the ideal-field experiment, and one public projective-slice trial has the stronger per-trial regularization bounds in Corollary 4.12. For concrete seeded keys, Proposition 2.2 adds the rejection-sampling and expansion-transcript terms. The rejection term is numerically larger than the pure geometric failure only at Level I and is displayed separately as an artifact of the ideal-to-concrete transfer.

6.3 Parameter consequences

The two attack surfaces constrain different parameters. Increasing only $V = v/3$ raises the projective-slice solve but leaves the base-field pseudo-oil terminal dimension essentially unchanged. Increasing only m raises the pseudo-oil residual but leaves the leading 2^V path count of the sliced key-recovery solve unchanged, although it improves the key-density exponents through $e = m - V$. A revised parameter search must therefore optimize both surfaces and charge the subspace extraction as well as the terminal solve. Any replacement set also requires a fresh analysis of key size, signing failure, and every attack already listed by the specification.

A structural countermeasure would have to remove or substantially alter the public extension-field UOV tuple used by projective slicing. Merely changing the public oil chart or concealing coordinate directions does not help: the new attack samples arbitrary projective planes and recovers the oil space from an ambient derivative kernel. Parameter inflation remains the immediate response within the current design.

6.4 Asymptotic, not practical

One degree-2^V quotient vector over the relevant auxiliary field already occupies approximately 0.102 EiB, 2.23×10^6 EiB, and 1.84×10^{14} EiB at Levels I, III, and V; materializing all coordinate polynomials is larger by another factor of roughly V. No implementation in this work approaches the full-parameter time or memory. The result is a parameter shortfall in the specification's asymptotic attack model, not a practical key recovery.

7 Conclusion

A public random projective V-plane necessarily contains an oil point. When a random square core on that plane is smooth, the point is unique and its ambient derivative kernel is the full hidden oil space. The full discriminant and the off-oil incidence bound make this an explicit almost-all-key theorem, with ideal-field bad-key probabilities below $2^{-364.5}$, $2^{-457.9}$, and $2^{-1038.0}$ and expected uniform indicators $124.5, 177.8, 235.4$.

The arbitrary-core method remains only as an incomparable-coverage robustness result, while the independent pseudo-oil path is conditional on its published extraction heuristic. No full-parameter computation is claimed; the conclusion is a substantial parameter shortfall in the asymptotic algebraic cost model used by the specification.

References

[1] H. Furue, Y. Ikematsu, F. Hoshino, T. Takagi, H. Kosuge, K. Yamakoshi, R. Akiyama, S. Nakamura, S. Orihara, and K. Kinjo. *QR-UOV Specification Document (Round 2 submission)*. NIST Additional Digital Signatures Project, 5 February 2025.

[2] QR-UOV team. *Round 2 Reference Implementation of QR-UOV*. Public source repository, December 2024.

[3] A. Kipnis, J. Patarin, and L. Goubin. Unbalanced Oil and Vinegar signature schemes. In *EUROCRYPT 1999*, pp. 206–222, 1999.

[4] H. Furue, Y. Ikematsu, Y. Kiyomura, and T. Takagi. A new variant of unbalanced oil and vinegar using quotient ring: QR-UOV. In *ASIACRYPT 2021, Part IV*, pp. 187–217, 2021.

[5] National Institute of Standards and Technology. Nine candidates advance to the third round of the Additional Digital Signatures for the PQC standardization process. NIST CSRC, 14 May 2026.

[6] G. Alagic, M. Bros, P. Ciadoux, et al. *Status Report on the Second Round of the Additional Digital Signature Schemes for the NIST Post-Quantum Cryptography Standardization Process*. NISTIR 8610, May 2026.

[7] O. Benoist. Degrés d'homogénéité de l'ensemble des intersections complètes singulières. *Annales de l'Institut Fourier*, 62(3):1189–1214, 2012.

[8] J. van der Hoeven and G. Lecerf. On the complexity exponent of polynomial system solving. *Foundations of Computational Mathematics*, 21(1):1–57, 2021.

[9] M. Safey El Din and E. Schost. Bit complexity for multi-homogeneous polynomial system solving—application to polynomial minimization. *Journal of Symbolic Computation*, 87:176–206, 2018.

[10] E. Schost. Computing parametric geometric resolutions. *Applicable Algebra in Engineering, Communication and Computing*, 13(5):349–393, 2003.

[11] T. X. Vu. A symbolic homotopy algorithm for solving composable polynomial systems. In *ISSAC 2026*, ACM, 2026; arXiv:2605.22514.

[12] D. Harvey, J. van der Hoeven, and G. Lecerf. Faster polynomial multiplication over finite fields. *Journal of the ACM*, 63(6):52:1–52:23, 2017.

[13] N. Pippenger and M. J. Fischer. Relations among complexity measures. *Journal of the ACM*, 26(2):361–381, 1979.

[14] V. Strassen. Gaussian elimination is not optimal. *Numerische Mathematik*, 13:354–356, 1969.

[15] W. Beullens. Improved cryptanalysis of UOV and Rainbow. In *EUROCRYPT 2021, Part I*, pp. 348–373, 2021.

[16] H. Furue and Y. Ikematsu. A new security analysis against MAYO and QR-UOV using rectangular MinRank attack. In *IWSEC 2023*, pp. 101–116, 2023.

[17] T. Suzuki, H. Furue, T. Ito, S. Nakamura, and S. Uchiyama. An extended rectangular MinRank attack against UOV and its variants. Cryptology ePrint Archive, Report 2025/739, 2025.

[18] P. Pébereau. One vector to rule them all: Key recovery from one vector in UOV schemes. In *PQCrypto 2024*, pp. 92–108, 2024; ePrint 2023/1131.

[19] A. May, M. Ostuzzi, and H. Ressler. Just Guess: Improved (quantum) algorithm for the underdetermined MQ problem. In *EUROCRYPT 2026, Part IV*, pp. 334–362, 2026; ePrint 2025/1788.

[20] M. Ostuzzi. Pseudo-oil subspaces and the geometry of underdetermined MQ problems. Cryptology ePrint Archive, Report 2026/1122, 2026.

[21] Y. Jin, Y. Pan, X. He, B. Gong, and J. Ding. Security analysis on UOV families with odd characteristics: Using symmetric algebra. In *PKC 2026, Part I*, pp. 428–454, 2026; ePrint 2025/1137.

[22] J. Thaler, K. Zhang, S. Kominers, and Q. Dao. *Polar Flags and Hodge Contractions*. Concurrent manuscript, 2026.

[23] J. Thaler, K. Zhang, S. Kominers, and Q. Dao. *Symmetry-Quotient Forgery Attacks on Odd-Characteristic SNOVA*. Concurrent manuscript, 2026.

[24] T. Aulbach, F. Campos, and J. Krämer. SoK: On the physical security of UOV-based signature schemes. In *PQCrypto 2025*, pp. 199–231, 2025; ePrint 2024/1818.

[25] J. Heintz. Definability and fast quantifier elimination in algebraically closed fields. *Theoretical Computer Science*, 24:239–277, 1983.

[26] G. Lachaud and R. Rolland. On the number of points of algebraic sets over finite fields. *Journal of Pure and Applied Algebra*, 219(11):5117–5136, 2015.

A Attack algorithms

Fast projective-slice equivalent-key recovery.

1. Convert the compressed public key to the specification's extension-field tuple over $K = \mathbb{F}_{127^3}$ and construct the fast solver's working extension L.
2. Sample $M \in L^{(V+O)\times(V+1)}$, reject unless rank $M = V+1$, and form all restricted quadrics $q_i(t) = t^\mathsf{T} M^\mathsf{T} P_i M t$.
3. Sample $R \in L^{V\times m}$, reject unless rank $R = V$, form $g = Rq$, and run a bounded fast-solver invocation; abort on a failed inversion, missing a posteriori certificate, or budget exhaustion.
4. Extract the L-rational roots of the geometric resolution, impose every unrecombined restricted equation, and discard nonroots.
5. For each survivor $x = Mt$, compute $\ker(P_1 x \ \cdots \ P_m x)^\mathsf{T}$. Require dimension O, Frobenius descent to K, total isotropy, and every public graph identity.
6. Apply the specification's public inverse pull-back to a verified oil space. On failure, resample (M, R).

Pseudo-oil direct forgery.

1. Expand the full base-field verification map as m homogeneous quadrics in $n = v + m$ variables over $\mathbb{F}_{127}$.
2. Run the enriched subspace extraction with $p = 2$, $(b_1, b_2) = (1, 1)$, $\ell_{\mathrm{po}} = 1$, and $k = 0$.
3. Restrict to the resulting m-dimensional pseudo-oil space.
4. Solve equation one in the coordinate of $\widetilde{U}$, then equation two in the coordinate of U_1, backtracking over their constant-size root tree.
5. Use equation three to eliminate one U_0 coordinate and solve the remaining $(m-3)$-variable square system.
6. Reverse the affine substitutions and verify the reconstructed signature against the original public map.

B Arbitrary-core fallback

The main attack uses a smooth complete intersection so that the fastest Kronecker theorem applies. The fallback instead targets only the nonsingular isolated roots of an arbitrary square core

$$Z_{\mathrm{ns}}(f) = \{y : f(y) = 0,\ \det J_f(y) \neq 0\}.$$

A full-rank planted derivative guarantees that the planted root belongs to this set, regardless of every other component and every prefix ideal. This appendix records the finite-characteristic specialization; its success condition is incomparable with the main theorem's, and its bad-key probability is smaller.

B.1 Finite-characteristic start fiber and separator

Lemma B.1 (Vandermonde product start system). *Let K' contain $2V$ distinct elements $a_1, \dots, a_{2V}$. Put*

$$L_a(X) = X_1 + aX_2 + \cdots + a^{V-1}X_V + a^V, \qquad g_i = L_{a_{2i-1}}L_{a_{2i}}.$$

Then $g = 0$ has exactly $C = 2^V$ affine roots over K', every root is simple, and a geometric resolution can be constructed in $\widetilde{O}(VC)$ operations.

Proof. Choose one node from each pair. The selected V linear equations have an invertible Vandermonde matrix and hence one solution; its monic polynomial has exactly the selected nodes as roots, so no unselected factor vanishes. Different selections give different roots, and the Jacobian is an invertible Vandermonde matrix times a nonzero diagonal matrix. Gray-code enumeration plus product-tree interpolation gives the stated cost. □

Lemma B.2 (Full random separator). *If the start and target nonsingular fibers each have degree at most C, all well-separating conditions used by the symbolic homotopy can be expressed as nonvanishing of a random linear form on at most C^2 nonzero vectors. A full random linear form over K' is therefore good with probability at least $1 - C^2/|K'|$.*

Proof. There are at most $\binom{C}{2}$ start-point differences, $\binom{C}{2}$ target-limit differences, and C nonzero leading-coefficient vectors, totaling C^2. A uniform linear form annihilates each fixed nonzero vector with probability at most $1/|K'|$. □

B.2 Characteristic-independent local updates

Lemma B.3 (Newton–Hensel doubling). *Let A be a commutative ring, $F \in A[T, X_1, \dots, X_V]^V$, and work modulo T^{2s}. If*

$$F(X) \equiv 0 \pmod{T^s} \quad \textit{and} \quad J_F(X) \textit{ is invertible modulo } T^{2s},$$

then

$$X^+ = X - J_F(X)^{-1}F(X) \pmod{T^{2s}}$$

satisfies $F(X^+) \equiv 0 \pmod{T^{2s}}$.

Proof. The correction is divisible by T^s. Taylor expansion cancels the constant and linear terms, while every remaining term contains two corrections and is divisible by T^{2s}. No factorial or characteristic-dependent denominator occurs. □

Lemma B.4 (Primitive-element renormalization). *Let $B = A[T]/(T^{2s})$, let $Q \in B[U]$ be monic, and represent coordinates X_i modulo Q. Suppose*

$$\delta(U) = \mathrm{rem}_U(\lambda(X) - U, Q) \in T^s B[U]$$

and write $Q'(U)\delta(U) = H(U)Q(U) + R(U)$ with $\deg_U R < \deg Q$. Define

$$Q^* = Q - R, \qquad X_i^* = \mathrm{rem}_U(X_i - X_i'\delta, Q^*).$$

Then $U \mapsto U + \delta(U)$ gives an isomorphism $B[U]/(Q^*) \to B[U]/(Q)$, with inverse $U \mapsto U - \delta(U)$; it sends X^* to X and makes $\lambda(X^*) = U$. Polynomial relations satisfied by X are preserved.

Proof. Every coefficient of δ, H, R is divisible by T^s, so products of two such quantities vanish modulo T^{2s}. First-order Taylor expansion gives

$$P(U \pm \delta) = P(U) \pm P'(U)\delta$$

for every polynomial P. Substituting $U + \delta$ into $Q^* = Q - R$ and using $Q'\delta = HQ + R$ shows that Q^* maps to zero modulo Q; the reverse substitution similarly maps Q to zero modulo Q^*. The two substitutions are inverse because $\delta(U + \delta) = \delta(U)$. The coordinate and primitive-element identities follow from the same first-order expansion, and an isomorphism preserves all polynomial relations. □

Theorem B.5 (Finite-characteristic nonsingular-root solver). *Let f be any V affine quadrics over a finite field of odd characteristic and put $C = 2^V$. Over an extension K'/K containing $2V$ distinct elements, replace the published product start system and integer-indexed separator in the Safey El Din–Schost nonsingular-solutions algorithm by Lemmas B.1 and B.2, and use the local updates above. The output is always a subset of $Z_{\mathrm{ns}}(f)$ and contains every nonsingular isolated root with probability at least*

$$1 - \frac{C^2}{|K'|}.$$

In particular, $|K'| \geq 8C^2$ gives success probability at least $7/8$.

Proof. Safey El Din–Schost's Proposition 5 assumes that the characteristic is at least the maximum of two explicit quantities. Their proof uses these two hypotheses separately. The first is the hypothesis of their Lemma 7 and is used to make the integer-indexed affine factors in the start system distinct; Lemma B.1 supplies a simple start fiber of the same degree without that restriction. The second is the hypothesis of their Lemma 14 and is used only to guarantee a sufficiently large integer-indexed separator family; Lemma B.2 replaces it by a full random linear form and the same C^2 bad-vector count.

The intervening target-fiber analysis explicitly passes to generalized power series in arbitrary characteristic. After the two replacements, the published Lemma 13 proof reuses the same global lift, target specialization, squarefree extraction, and Jacobian Clean steps; its degree and precision quantities depend only on the unchanged start and target degrees. The local Newton and primitive-parameter identities are supplied by Lemmas B.3 and B.4. Thus the proof of Proposition 5 carries over with failure probability $C^2/|K'|$. Remark 15 and the final Clean call give one-sidedness: a bad separator may omit roots or report failure but cannot output a point outside $Z_{\mathrm{ns}}(f)$ [9, Lemmas 7, 13, 14, Proposition 5, and Remark 15]. □

B.3 A coordinate-slice batch

The fallback's coverage depends on how many of the O coordinate directions give a nonsingular planted slice. For $1 \le j \le O$, fixing the final public coordinates to e_j leaves a V-variable affine system with planted point $(-Ge_j, e_j)$. In central coordinates and after translation it is

$$f_{i,j}(y) = y^{\mathsf{T}} A_i y + 2y^{\mathsf{T}} B_i e_j.$$

Let J_j be the derivative of all m equations at the planted point.

Theorem B.6 (Full-derivative coordinate batch)**.** *In the ideal-field experiment, $J_1, \dots, J_O$ are independent uniform $m \times V$ matrices. Put*

$$\tau_{m,V} = 1 - \prod_{r=0}^{V-1} (1 - Q^{r-m}).$$

The exact probability that every direction has deficient full derivative is $\tau_{m,V}^O$. Its base-two logarithms at Levels I, III, and V are

$$-1132.17, \qquad -1635.35, \qquad -2935.25.$$

Proof. The ith row of J_j is $2(B_i e_j)^{\mathsf{T}}$. For fixed i, these are the independent columns of the uniform matrix B_i; independence over i completes the claim. □

B.4 Dense-quadratic batching and costs

At precision s, the lift works in a coefficient algebra

$$\mathcal{A}_s = K'[U,T]/(q(U), T^s).$$

Flatten the V quadratic Hessians into a $V^2 \times V$ matrix and the lifted coordinates into a $V \times \dim \mathcal{A}_s$ table. Their product forms every Jacobian coefficient simultaneously. Any bilinear matrix-multiplication identity over K' remains valid over the associative K'-algebra $\mathcal{A}_s$. The lift carries the inverse Jacobian from the simple start fiber and updates it by Newton multiplication, so no division by an arbitrary coefficient-algebra element is introduced. Once the Jacobian contractions are known, quadratic values follow from

$$f_i(X) = \frac{1}{2} X^{\mathsf{T}} J_i(X) + \frac{1}{2} b_i^{\mathsf{T}} X + c_i.$$

Theorem B.7 (Arbitrary-core lifting bound)**.** *With $C = 2^V$ and target precision $C_0 = 2^{V-1}(V+2)$, the finite-characteristic solver of Theorem B.5 can be implemented in*

$$\widetilde{O}\big((V^\omega + V^2) M(C) M(C_0)\big) = \widetilde{O}(V^{\omega+1} 4^V)$$

operations in K'.

Proof. At precision s, coefficient batching computes all Jacobian contractions in $\widetilde{O}(V^\omega M(C) M(s))$ operations. The quadratic-value identity above costs $O(V^2 M(C) M(s))$, and the inverse-Jacobian, coordinate-correction, and primitive-parameter updates fit the same bound. The precision doubles geometrically, so superlinearity of polynomial multiplication bounds the sum by the final stage $s = C_0$. Start parametrization, target specialization, squarefree extraction, gcd operations, and Clean are polynomial in V times C and the dense quadratic evaluation length and are dominated by the displayed 4^V term. Since $M(C) = \widetilde{O}(2^V)$ and $M(C_0) = \widetilde{O}(V 2^V)$, the second form follows; see also the finite-field multiplication bounds of Harvey–van der Hoeven–Lecerf [12]. □

Combining this solver with the coordinate batch in Theorem B.6 gives the following robust line. The Strassen column is field-uniform and is regenerated by the accompanying script directly from Theorem B.7. Replacing Strassen by ordinary cubic matrix algebra gives the final uniform column.

Table 4: Arbitrary-core fallback. The Strassen entry is the direct indicator with the uniform indicator in parentheses; the cubic entry is the uniform indicator obtained from the same formula with $\omega = 3$.

Level	Field-uniform Strassen	Cubic uniform	Target
I	131.79 (138.83)	139.91	143
III	182.26 (189.77)	190.96	207
V	236.18 (244.06)	245.34	272

Let G be the number of full-rank coordinate directions. In the ideal-field experiment $G \sim \mathrm{Bin}(O, 1 - \tau_{m,V})$. Conditional on $G > 0$, one trial succeeds in choosing a regular direction, an invertible square recombination, and a good separator with probability at least

$$\frac{G}{O}\left(\prod_{r=0}^{V-1}(1 - Q^{r-V})\right)\frac{7}{8}.$$

Using $1/G \le 2/(G+1)$ and

$$\mathbb{E}\left[\frac{1}{G+1}\right] = \frac{1 - \tau_{m,V}^{O+1}}{(O+1)(1 - \tau_{m,V})}$$

shows that the expected number of full solver calls, averaged over ideal-field keys conditional on $G > 0$, is below $16/7 + o(1)$. The exact all-directions-deficient probability is $\tau_{m,V}^{O}$ by Theorem B.6; fixed-budget concrete success transfers by Proposition 2.2. On every key with $G > 0$, repeated public trials are Las Vegas after substitution, graph completion, isotropy, and inverse-pull-back verification; on an all-deficient key the procedure may fail to terminate.

The fallback's uniform costs are $14.36, 11.97, 8.64$ bits above the fast projective line, while its explicit ideal-field bad-key probability is much smaller; the two orderings are incomparable. It therefore serves as a robustness line rather than the headline attack.

C Negative attack audit

Why the old one-slice exceptional-locus program was abandoned. For a fixed planted slice, the exceptional locus contains the determinantal set on which the full planted derivative has rank below V: every recombination is then singular at the planted point. This component has codimension $m - V + 1 = 3, 3, 4$ and exact ideal-field density $\tau_{m,V}$, about $2^{-62.90}, 2^{-62.90}, 2^{-83.86}$. Thus a category-negligible density theorem for one unconditioned slice is false. The projective attack changes the randomization space and proves a different joint exceptional-set bound.

All graph columns at once. The full coupled graph system is heavily overdetermined, but it is not a random overdefined MQ instance. Once one graph column is known, all others are linear. The global path-count theorem Theorem 3.6 shows that every nonzero multihomogeneous Bézout count for a square selection is at least 2^V, attained by one-column recovery and tree-structured completion. Equation selection alone therefore does not beat the main square solve.

Using all spare slice equations directly. The fixed affine systems have shapes 54-in-52, 78-in-76, and 105-in-102. Their standard semiregular Hilbert-series screens first become nonpositive at degrees $25, 36, 46$, with logarithmic monomial counts $66.653, 97.829, 128.506$. Even an optimistic quadratic sparse-linear-algebra charge is above the fast discriminant line before field arithmetic, matrix construction, memory, and circuit conversion.

Conditioning and low-degree lifts. Exact affine Macaulay calculations through six variables agree with matched random controls at every preterminal degree; the planted point changes only the final one-dimensional quotient. Higher tensor and moment lifts contain exact exterior-power kernels, but their row counts substantially overpredict rank because of Macaulay syzygies. No stable recursive radical, module, or low-rank decomposition was found in the included small-dimensional controls. These tests rule out the implemented shortcuts, not all conceivable algorithms.

D A determinant-pencil characterization

We record a structural observation linking common quadric roots to the singular locus of the determinant pencil. It clarifies the geometry but did not lead to a faster attack.

Let $P_1, \dots, P_m \in \mathrm{Sym}_N(K)$ be homogeneous sliced matrices and define

$$M(\lambda) = \sum_i \lambda_i P_i, \qquad \Delta(\lambda) = \det M(\lambda).$$

For a common root x, put $K_x = \ker[P_1 x \ \cdots \ P_m x]$.

Proposition D.1 (Critical-locus duality)**.** *The projective space $\mathbb{P}(K_x)$ is contained in the singular locus of $\Delta = 0$. Conversely, every corank-one critical pencil member has a kernel vector that is a common quadric root.*

Proof. For $\lambda \in K_x$, $M(\lambda)x = 0$. At corank one, $\operatorname{adj} M(\lambda) = cxx^\mathsf{T}$, so Jacobi's identity gives

$$\frac{\partial \Delta}{\partial \lambda_i} = \mathrm{tr}(\operatorname{adj} M(\lambda) P_i) = cx^\mathsf{T} P_i x = 0.$$

At corank at least two, the adjugate is zero. Conversely, if a critical member has corank one with kernel y, the same identity gives $y^\mathsf{T} P_i y = 0$ for every i. □

Every corank-at-least-two member is automatically critical and creates a large extraneous component. Exact finite-field tests confirm the characterization.

E Reproducibility and scope

The source bundle contains the manuscript, the exact parameter, probability, and cost script, and the exact small-field validation. The numerical script regenerates the key-bad, one-trial, two-trial, fast, fallback-Strassen, fallback-cubic, and pseudo-oil-Strassen entries retained in this version. The validation script checks geometric smoothness chart by chart and verifies derivative-kernel oil recovery on every smooth toy core.

Experimental results support only the algebraic identities they test; they are not extrapolated into production runtimes.

Exact results. The public extension-field reduction; ideal-field distribution; projective-oil intersection; joint-discriminant nonvanishing outside the three explicit loci; finite-field incidence and degree bound; full-core solver regularity; derivative-kernel recovery and descent; one-column completion; oil-space uniqueness codimension; multihomogeneous path floor; finite-characteristic fallback specialization; and the pseudo-oil residual reduction after a valid decomposition.

Distributional or computational-transfer results. The key-density and coordinate-batch probabilities use the ideal-field experiment. Concrete fixed-budget verified-success experiments transfer through Proposition 2.2. The fast theorem excludes a bounded ideal-field set.

Heuristic or asymptotic components. Pseudo-oil existence, extraction, and branch behavior; soft-O polynomial and matrix arithmetic; and all production cost estimates. The arbitrary-core solver does not use a global geometry heuristic, but its dense-quadratic cost is asymptotic.

Outside scope. The artifact does not provide a byte-exact official compressed-key KAT attack, a full-parameter execution, a manageable-memory geometric-resolution implementation, calibrated constants for fast matrix multiplication, or a theorem for malformed public keys.